GolfIt

Group 4: Sultans of Swing
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1. Game Vision

GolfIt is a real-time multiplayer arcade game that combines the precision of classic minigolf with a dynamic, interactive environment. Unlike traditional turn-based golf, players compete simultaneously in a race against time and each other to finish the course in the fewest strokes possible.

The primary "hook" is the **interactive map environment**. Players don't just focus on their own ball; they can trigger environmental changes (like moving bridges or closing gates) that affect their opponents' paths. This creates a strategic layer where "sabotaging" a faster player becomes a viable tactic to stay in the lead.

2. Objectives and Win/Loss Conditions

- **Primary Goal:** Sink the golf ball into the hole using the lowest number of strokes (moves) possible.
- **Win Conditions:** A match consists of several rounds (different maps). The winner is the player with the lowest cumulative score (total strokes) across all rounds.
- **Loss Conditions:** If a player fails to complete a hole within the designated time limit or exceeds the maximum stroke count (e.g., 12 strokes), they receive a "DNF" (Did Not Finish) penalty score, significantly hindering their chances of winning the overall match.

3. Game Rules

1. **Setup:** All players spawn simultaneously at the starting tee. Each player controls a ball with a unique color or skin to ensure visibility during the game.
2. **Turn/Flow Structure:** The game operates in **real time**. There is no waiting for opponents; players can kick their ball as soon as it comes to a complete stop from the previous move. Ball positions and movement animations are synchronized across all clients via the server.
3. **Constraints:** Balls cannot pass through solid walls or closed gates. Contact with walls results in a physical bounce based on the angle of incidence. Falling off the map (if applicable) resets the ball to its last stable position with a one-stroke penalty.

4. Core Mechanics

- **Movement & Interaction:** Players click and drag on their ball to set the direction and power of the kick. A visual "Power Bar" (changing from green to red) indicates the intensity, while an arrow overlay shows the projected trajectory.
- **Environmental Interaction:** Maps contain "Trigger Buttons." When a ball rolls over a button, it activates a mechanism: opening a shortcut, rotating a windmill, or closing a door in front of the hole.

5. Networking & Multi-User Interaction

- **Server Responsibilities:** The server communicates ball movements to all players, manages the countdown timer for each round, and broadcasts the state of environmental triggers to all clients.
- **Client Responsibilities:** The client handles the GUI rendering, captures the player's drag-and-drop input for kicks.
- **Interaction:** A shared lobby allows players to join before the match starts. A global chat is available for both the lobby and the in-game HUD to allow for "friendly" chat during the game.

6. User Interface (UI) and Mockup Overview

- **Main Game View:** A **top-down perspective** of the minigolf course, showing all active players' balls and interactive environmental elements.
- **Heads-Up Display (HUD):** A clean overlay showing the current stroke count, the remaining time for the round, and a leaderboard showing the rank of all players in real-time.
- **Social/Technical Elements:** A collapsible chat window on the side.

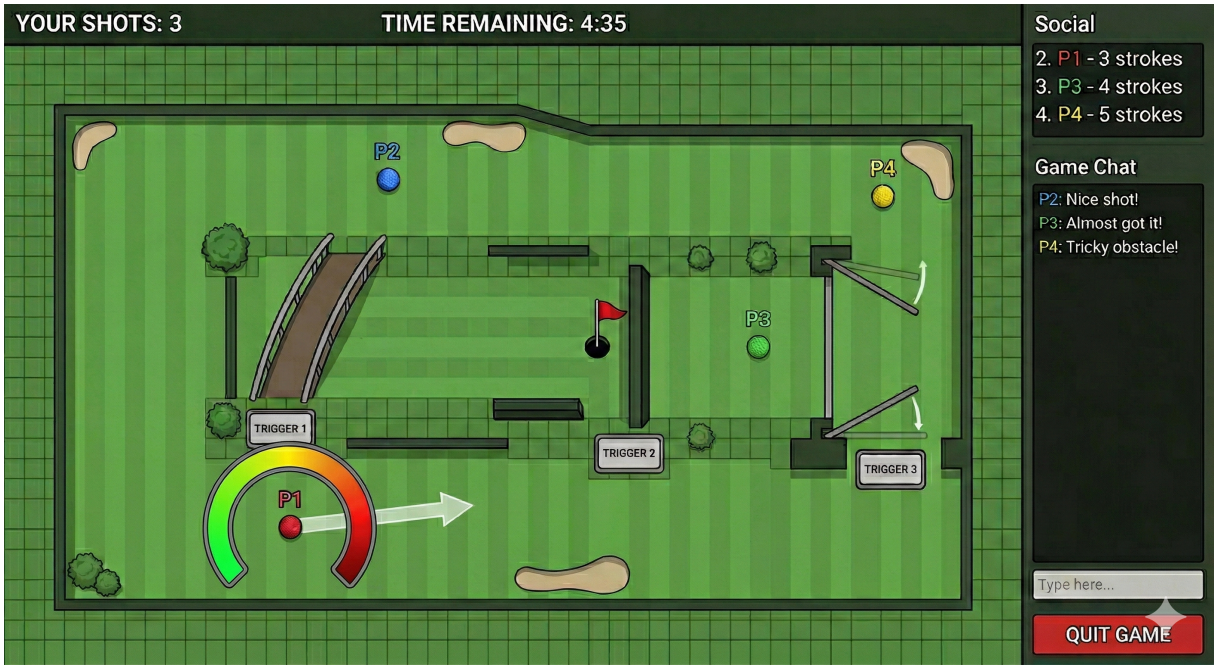


Figure 1: Mockup gameboard, generated with Nano Banana 2